

Graphs of functions - Set 2 - Solutions

Specification MM8.1, MM8.5-MM8.7, M4.15.

Answer key

Question	Answer	Topic
1	A	Graphs of functions
2	A	Graphs of functions
3	E	Graphs of functions
4	F	Graphs of functions
5	B	Graphs of functions
6	G	Graphs of functions

Worked solutions

Question 1 GRA-11 · Graphs of functions · spec MM8.7

The curves $y = 2^x$ and $y = x^2$ intersect at how many points?

Answer: A

They meet at $x = 2$ and $x = 4$ ($4 = 4$, $16 = 16$), and once for negative x (at $x = 0$: $1 > 0$; at $x = -1$: $\frac{1}{2} < 1$). For $x > 4$, 2^x grows faster. Three intersections.

Question 2 GRA-13 · Graphs of functions · spec MM8.7

For how many values of k does the equation $|x^2 - 4| = k$ have exactly three real solutions?

Answer: A

The graph of $y = |x^2 - 4|$ is a W-shape with minima 0 at $x = \pm 2$ and a local maximum 4 at $x = 0$. A horizontal line gives three solutions only when it passes through the local maximum: $k = 4$. One value.

Question 3 GRA-14 · Graphs of functions · spec MM8.1

The graph of $y = \sqrt{x}$ is transformed to $y = \sqrt{4 - x}$. Which sequence of transformations achieves this?

Answer: E

$\sqrt{4 - x} = \sqrt{-(x - 4)}$: reflect $y = \sqrt{x}$ in the y -axis to get $\sqrt{-x}$, then translate 4 to the right. (Translating first then reflecting gives $\sqrt{-x - 4}$.)

Question 4 GRA-15 · Graphs of functions · spec MM8.6

The cubic $y = x^3 + ax + b$ has three distinct real roots. Which of the following statements must be true?

- I $a < 0$
- II $b \neq 0$
- III The curve has two turning points.

Answer: F

Three distinct real roots require two turning points (III), which needs $y' = 3x^2 + a$ to have two roots, so $a < 0$ (I). II is false: $x^3 - x$ has roots 0, ± 1 and $b = 0$.

Question 5 X2-09 · Graphs of functions · spec MM8.7

The graphs of $y = f(x)$ and $y = g(x)$ meet at exactly three points. Which of the following statements must be true?

- I The equation $f(x) = g(x)$ has exactly three real solutions.
- II The graph of $y = f(x) - g(x)$ crosses the x -axis exactly three times.
- III $f(x) - g(x)$ is a polynomial of degree at least 3.

Answer: B

I is the definition of an intersection. II fails because the difference could touch the axis rather than cross it at one of the points. III fails because f and g need not be polynomials at all (e.g. $\sin x$ and a line).

Question 6 X2-21 · Graphs of functions · spec MM8.6

$p(x)$ is a polynomial of degree 4 with real coefficients. Which of the following statements must be true?

- I** $p(x) = 0$ has at least one real root.
- II** $p(x) = 0$ has at most four real roots.
- III** The graph of $y = p(x)$ has at least one turning point.

Answer: G

I fails: $x^4 + 1$ has no real roots. II: a degree-4 polynomial has at most 4 roots. III: an even-degree polynomial tends to the same infinity at both ends, so it must turn at least once.